References: **Lecture Slides for Lectures #1 & #2**
PDF Files on Moodle: ***Chart of Nuclides Key, Co-60 Activation Levels, Uranium Decay Series, Table 3-11 Tabulated Reactor Data, Properties of Elements - Lamarsh Table II.3; Select U-Np-Pu Isotopes.***
Read: **Reactor Theory Chapter #1: §1.1 – §1.3**

1. What are the following values for Cobalt (*see Chart of Nuclides key and ^{60}Co handout*)?
   a) The ^{59}Co resonance integral leading to the Isomeric State? ____________
   b) The ^{59}Co thermal neutron capture cross section leading to the Ground State?______________
   c) The half-life of the ^{60}Co Ground state = ___________________
   d) The half-life of the ^{60}Co Metastable (Isomeric) state = __________________
   e) The decay modes of the ^{60}Co Ground state = _____________________
   f) The decay modes of the ^{60}Co Metastable (Isomeric) state = ________________

2. Give an example of an isotope that is naturally occurring and radioactive: ________________

3. Does the cross section (i.e. probability) of undergoing an absorption reaction (σ_a) generally increase or decrease as a fast neutron moderates and loses energy?____________________

4. What is the decay chain leading from ^{234}U to ^{222}Rn?:________________________________

5. What are the *two types* of neutron absorption reactions? Give an example of each as they might occur inside a reactor core.

6. What is the difference between elastic and in-elastic neutron scattering?

7. Give examples of:
   a fertile nuclide:______________
   a fissile nuclide:______________
   a fissionable nuclide:______________

8. What is the total amount of energy released from the fission of a ^{235}U nucleus? Is more energy given off immediately (promptly) or delayed?

9. Reference Lamarsh Table II.3 – 'Properties of the Elements and Certain Molecule' and fill in the values in the tables below:

Which material has the best moderating properties?:

Material	Σ_a (cm^{-1})	Σ_s (cm^{-1})	Ratio: Σ_s/Σ_a
(Light) Water (H2O)			
Heavy Water (D2O)			
Beryllium			
Graphite			

Which material would make the best control rod absorber (based on these parameters alone)?

Material	Atom Density (N_a) (atoms/cm3)x10^{24}	Σ_a (cm^{-1})
Boron		
Cadmium		
Gadolinium		
Lead		

<u>See lab questions – next page!</u>

Lab Questions – PULSTAR Systems and Schematics

REF: Lab Handouts & Schematics

10. What are the following values and parameters for a 25 Assembly PULSTAR Core?

a) Fuel material & % Enrichment: ___________________________
b) Fuel Cladding material: _________________________
c) 5×5 Core Fuel mass = __________kg
d) Mass ^{235}U in fresh core = __________kg
e) Length of Active Fuel = __________inches
f) Moderator material: ______________________
g) Reflector material(s): _______________________________________
h) Control Rod Absorber material(s): ________________________________
i) Control Rod Clad material(s): ________________________________\

11. SCRAM Matching – match the SCRAM to the setpoint (*answers may be used more than once, or not at all…*)

Ans.		SCRAM
_____	1.	Linear Power Channel – Over Power SCRAM
_____	2.	Safety Power Channel – Over Power SCRAM
_____	3.	Log-N Channel – Flow/Flapper SCRAM Enable
_____	4.	Safety Power Channel – Flow/Flapper SCRAM Enable
_____	5.	T_2 Pool Temp
_____	6.	Pool Level
_____	7.	Primary Coolant Flow
_____	8.	Manual SCRAM

	Setpoint
A.	116°F
B.	150 kW
C.	2000°F
D.	-36 inches
F.	475 GPM
G.	Red Button
H.	1.20 MW
J.	2.05 MW

12. What are the trips associated with the Source Range Channel? ______________________________

13. Does the PULSTAR have a confinement or a containment?_________________________ What is its main purpose?

14. What are the three (3) main functions of the primary water?

15. What is the function of the three delay tanks in the PPV?

16. Which describes the correct flow path of heat from the reactor core to the atmosphere?
   a. Core, Primary Pump, ^{16}N Delay Tank, Heat Exchanger, Cooling Tower
   b. Core, Heat Exchanger, ^{16}N Delay Tank, Primary Pump, Cooling Tower
   c. Core, ^{16}N Delay Tank, Heat Exchanger, Primary Pump, Cooling Tower
   d. Core, ^{16}N Delay Tank, Primary Pump, Heat Exchanger, Cooling Tower